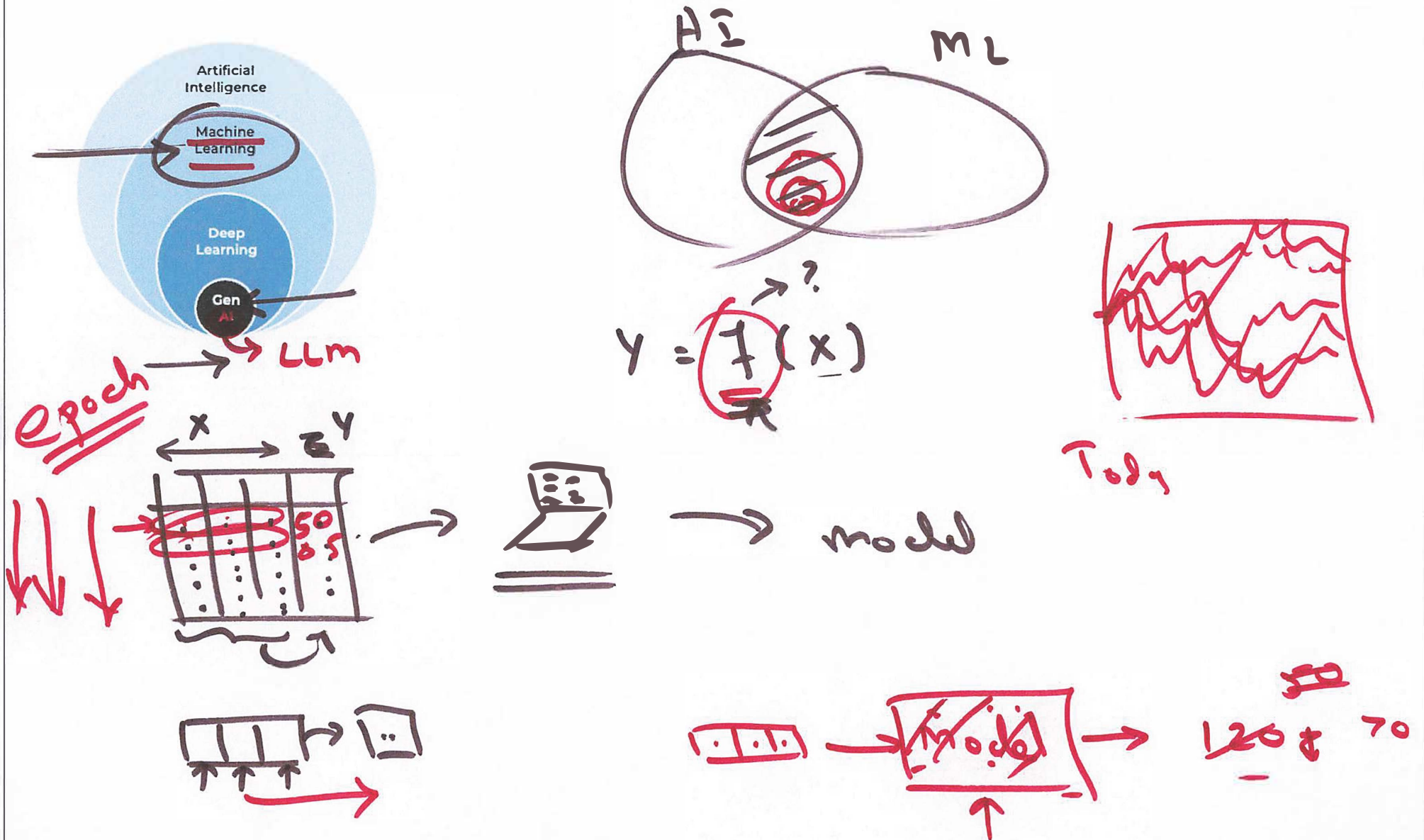
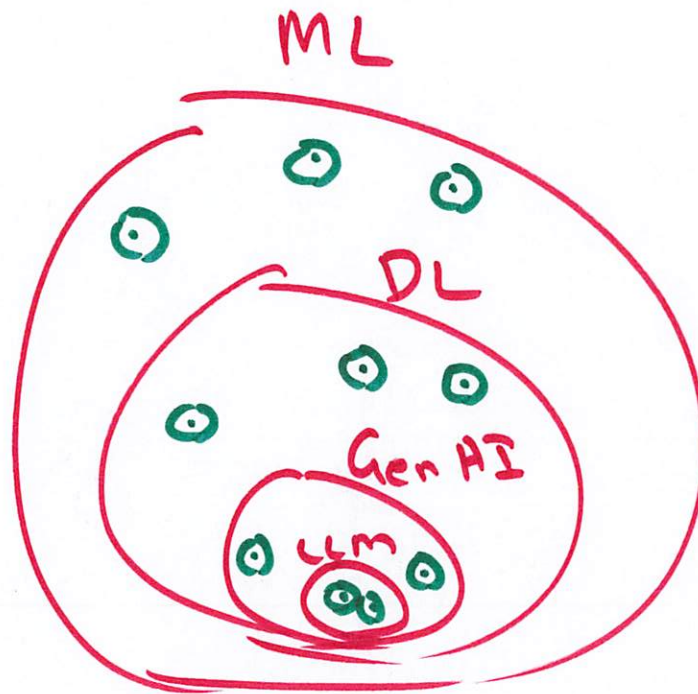


AI and Machine Learning

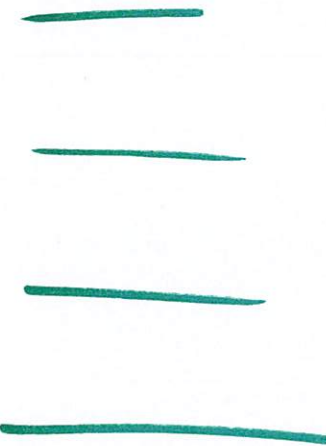




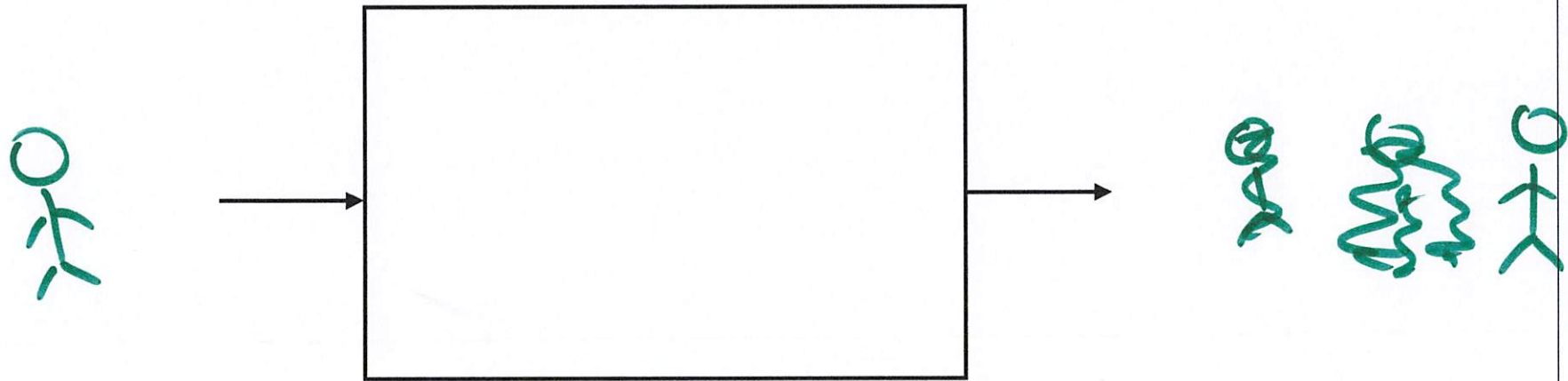
Retail

Health care

Tech.

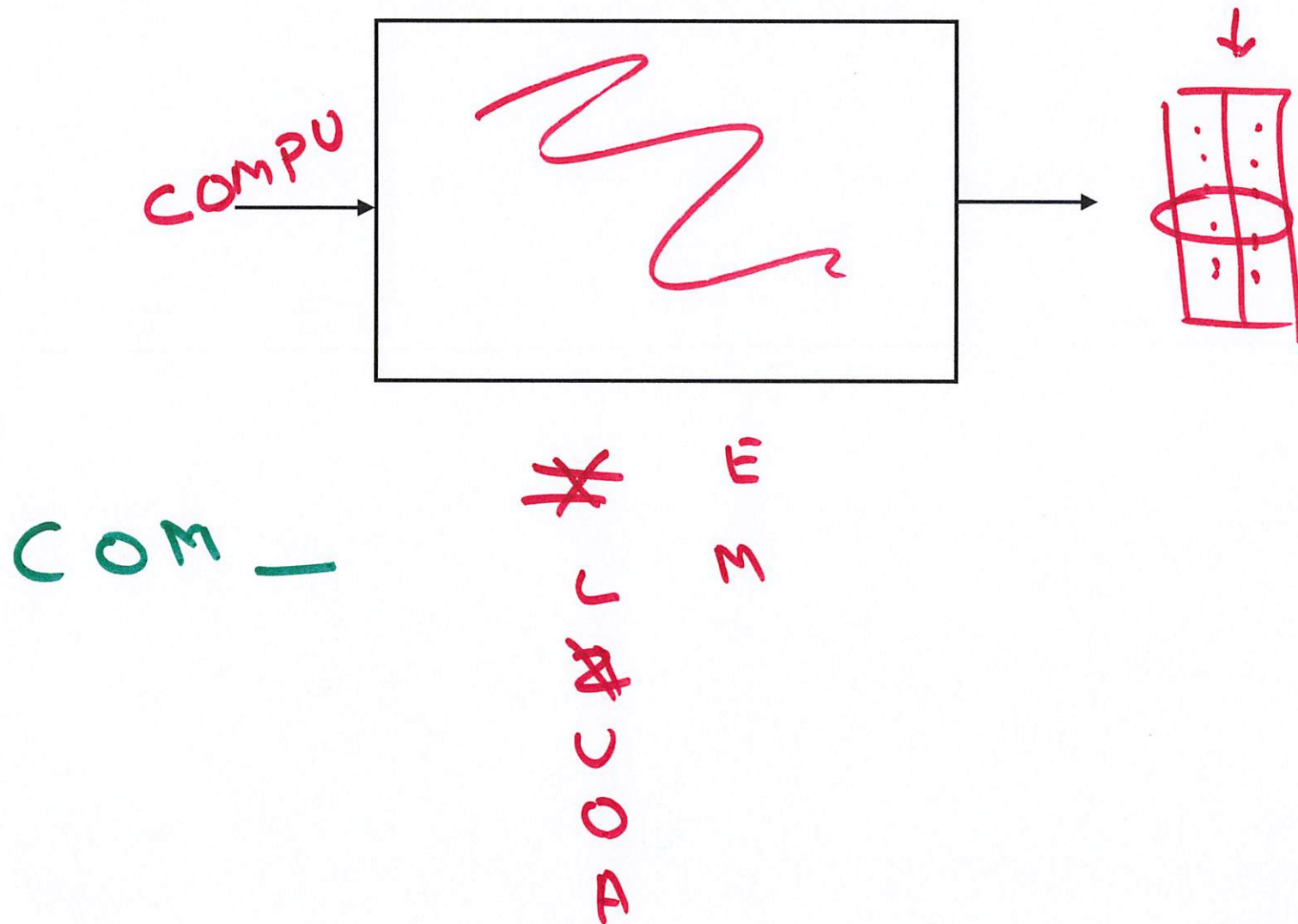


Learning a pattern and replicating it



Small Character Model !

- Learn patterns in character sequences and replicate that?

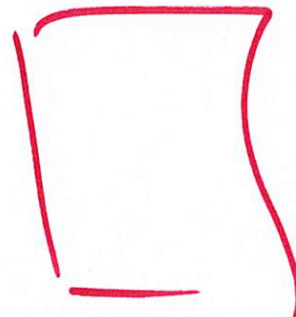


com P

comp I

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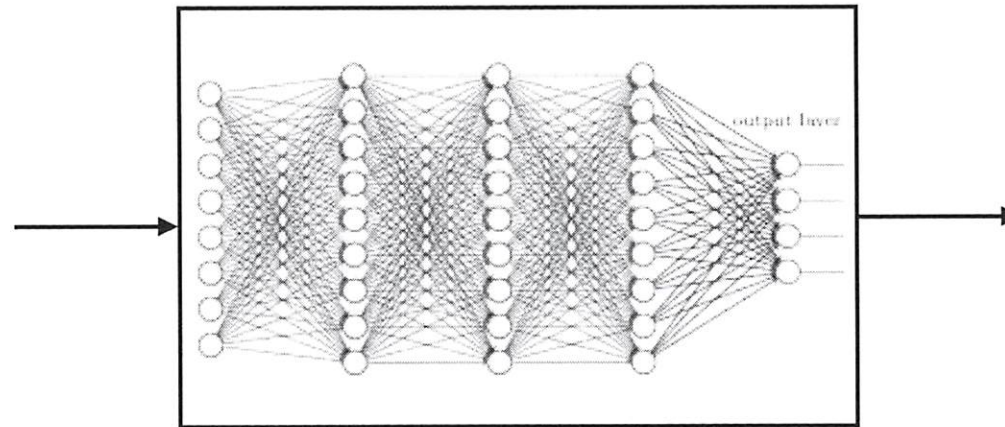
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L	<u>0.3</u>
O	<u>0.2</u>
A	<u>0.2</u>



E	<u>0.25</u>
M	<u>0.25</u>
P	<u>0.25</u>
R	<u>0.15</u>
B	<u>0.1</u>
.	<u>0</u>

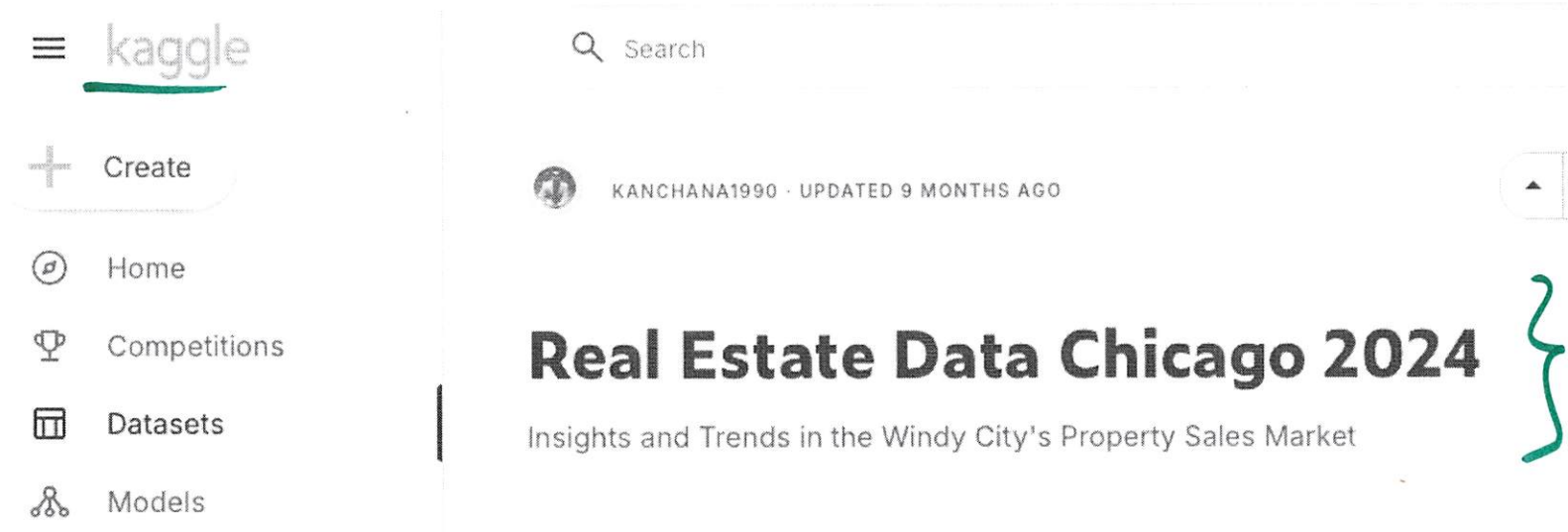
Large Language Models!

- Lets start with a very simple language model:



Large Language Models!

- Lets start with a very simple “small character model”
- Data



↓ ↓ ↓
Come check out this amazing, move-in ready, 5 bed, 3.5 bath in the Beautiful Edgebrook of Chicago. Completely updated within the past few years with attention to detail. Quality is everywhere throughout this home including a finished basement with a suite-like style bedroom, walk-in closet and it's own bathroom. Wake up with a cup of coffee off of your balcony connected to the master bedroom. Granite in the kitchen, gorgeous cabinets and high end appliances make this kitchen a dream. Amazing hardwood floors throughout the home, a spacious living room, large windows, 2 wood burning fire places and plenty of hot...

Total 2 Million Characters

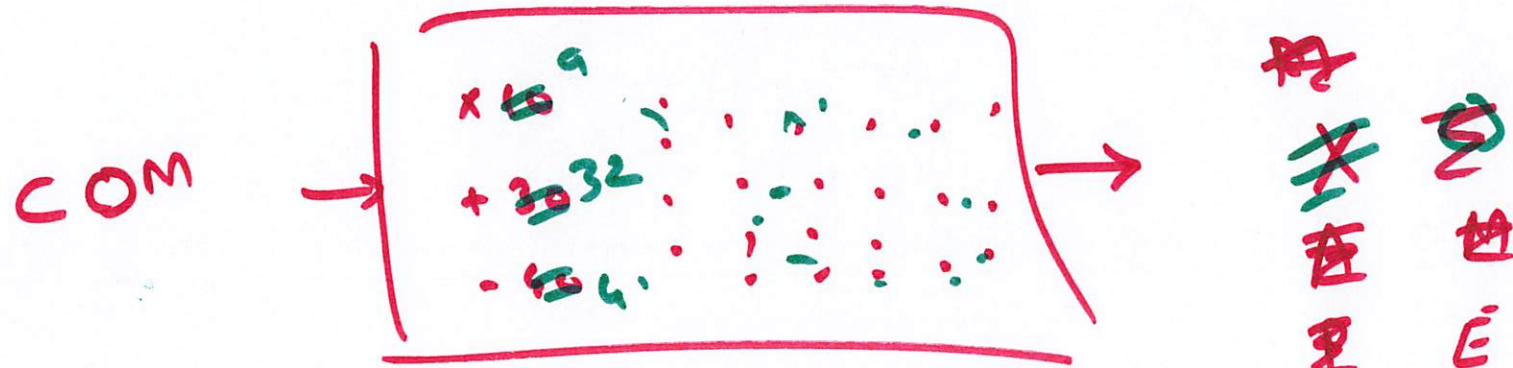
Come check out this amazing, move-in ready, 5 bed, 3.5 bath in the Beautiful Edgebrook of Chicago. Completely updated within the past few years with attention to detail. Quality is everywhere throughout this home including a finished basement with a suite-like style bedroom, walk-in closet and it's own bathroom. Wake up with a cup of coffee off of your balcony connected to the master bedroom. Granite in the kitchen, gorgeous cabinets and high end appliances make this kitchen a dream. Amazing hardwood floors throughout the home, a spacious living room, large windows, 2 wood burning fire places and plenty of hot water with two 40 gallon tanks. A large 52' x 110' lot giving this home a big and beautiful backyard. Top rated Edgebrook grammar school, walking distance to local shops and restaurants, and close to Caldwell woods that has awesome walking/bike trails. Don't forget about the Billy Caldwell golf course and easy access to 90/94 getting you to O'Hare airport or Downtown quickly. Showings start Monday 5/13.

Move in ready 2 story Mount Greenwood home on huge 49x134 lot, with extra wide driveway, perfect for parties. 2beds up and 2 beds down, plus fully finished basement. First floor features a large family room, beautiful formal dining with new south facing windows and galley kitchen. walk in pantry. 2 good size beds with fully built out closets and sliding glass door to rear deck. Upstairs features 2 huge bedrooms and full bath, new carpeting. Basement has beautiful family room and custom built bar. Overhead sewers ensure a dry basement. Half bath and plenty of storage. 2.5 car garage with epoxy flooring, and newer concrete driveway. Mount Greenwood public school district, close to Ridge Country Club.

Investor Special MONEY MAKING Gem. This is your chance to be the owner of this FULLY REMODELED Townhouse in South Chicago! House features brand new deck, new walls, new furnaces, new water heaters, new floors, brand new windows AND FINISHED BASEMENT. Each floor has a full kitchen as well.

Renovated Brick 1.5 story house with finished basement. Modern kitchen with high quality cabinetry, quartz countertops. New windows, appliances, Hardwood floors, water heater, windows. Two HVAC systems, second floor HVAC systems new, first floor HVAC systems about 4 years old. Garage roof new, house roof about 2 years old. Hug backyard. Convenient Location. 4 bedrooms, 2.1 bathrooms. 3 nice sized bedrooms on 2nd Level, 1 bedroom on main level. This home is in a prime location, offering easy access to Devon, Peterson, parks, award winning schools, eclectic restaurants, nature preserves, and major CTA transportation routes, including a new Metra Peterson line.

This almost 3000sq foot home was thoughtfully and lovingly renovated to retain the soul of this turn-of-the-century gem. Seamlessly blending its original character with carefully selected upgrades to honor its timeless design. Modern comforts harmonize with the historical integrity, ensuring this home retains its charm. The ceiling boasts rich wood beams that add a striking focal point, complemented by built-in cabinets featuring leaded glass doors for added elegance. Complete kitchen renovation with dark Maple cabinets and marble countertops to honor the old, paired with stainless steel



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~~Epoch~~

Ours vs GPT

	Ours	GPT
Type	Char Gen	Word Gen
Architecture	RNN	Transformer
Parameters	4 Million	2 Trillion
Training	<u>2</u> Million Chars	<u>10 Trillion Words</u>
Cost	<\$1	>\$150 Million
Comp Time	10 mins x 1 Comp	150 days x 10000 Comp